Dual-Route Sparse Evaluation for Grey Wolf Optimization: Which Candidates Deserve the Real Evaluations?

Draft – sparse_q research branch

***Abstract*—When objective evaluations are expensive, the real-evaluation budget - not arithmetic - decides solution quality, and the central question of a population method becomes *which candidates deserve a real evaluation*. This paper studies that question for the grey wolf optimizer (GWO) under a strict equal-evaluation protocol. Every generation generates $2N$ cheap candidates from two routes: an incumbent-refinement route (the standard GWO leader-aggregation moves plus success-scaled shell samples around the incumbent) and a verified multi-region route built from an elite archive of already-evaluated solutions (*verified* centres only; unevaluated candidates never influence leaders, rankings or archives). A fixed number K of the $2N$ candidates receive the real evaluations; the frozen variant uses $K = 10$ and uniform random selection over the pool. On CEC2017 at $D = 50$ with 100,000 true evaluations and 30 independent runs, the method reduces the median error ratio against standard GWO to 0.53 with 28 of 29 function wins, improves the 10%/20%-budget error ratios to 0.80/0.77, and reaches a statistical tie with the author-faithful I-GWO (median ratio 0.94, 17/12 function wins). A pre-registered ablation ladder answers the natural alternatives: directed screening by incumbent geometry, history-informed selection, fixed exploration quotas, randomly perturbed quotas, and three lightweight Q-learning quota schedulers were all tested; none of them beats the simple uniform policy, and the Q scheduler only outperforms *random* quota perturbation. The paper therefore contributes (i) a simple, evaluation-neutral sparse framework for GWO whose gains come from the candidate pool rather than from operator stacking, (ii) a complete, honest ablation of the screening and quota design space, and (iii) separate evidence for exploration efficiency and stagnation behaviour, with the non-supported parts reported explicitly.**

***Index Terms*—Grey wolf optimizer; sparse evaluation; expensive optimization; evaluation budget; Q-learning; CEC2017.**

I. Introduction

EVOLUTIONARY and swarm intelligence algorithms are increasingly used on problems where a single objective evaluation is expensive. In that regime the implicit design assumption of most population methods - evaluate every candidate of every generation - spends the scarcest resource on redundant confirmations. Sparse-evaluation frameworks such as spatial-screening PSO and sparse adaptive-quota CSO address this by evaluating only a subset of candidates per generation and letting the survivors carry the search. Grey wolf optimization (GWO), whose population moves by attraction to three leaders, has not been systematically studied under this lens: its update equations are simple, but its evaluation allocation is fixed - every wolf, every generation.

This paper asks a deliberately narrow question: *under an identical true-evaluation budget, which candidates should a wolf pack evaluate?* We build a minimal sparse framework in which each generation generates cheap candidates from two routes - refinement near the incumbent and verified multi-region proposals drawn from an archive of already-evaluated solutions - and spends the real evaluations by a simple selection policy. The resulting method, DRS-GWO (dual-route sparse GWO), changes neither the GWO update equations nor the budget accounting, and is therefore directly comparable with standard GWO and with strong baselines under equal true-evaluation budgets.

The contributions of this draft are:

- **A dual-route sparse-evaluation framework for GWO** in which unevaluated candidates never influence the search state (verified-state discipline), and the real evaluations of each generation are allocated by a single, parameter-free selection policy (uniform random over the pool).
- **A complete pre-registered ablation of the design space**: candidate-pool composition, evaluation density K, directed/informed screening, fixed/random quotas and three Q-learning quota schedulers. All negative results are retained; the study shows that in this setting the pool composition, not screening sophistication, carries the performance.
- **Separate, evidence-based conclusions for two objectives**: useful exploration efficiency (supported: earlier and larger error reductions and higher useful-novelty rates) and true stagnation recovery (reported separately, including the cases where the evidence is insufficient).

II. Related Work

A. Sparse and evaluation-efficient optimization

Sparse-evaluation frameworks evaluate only K of the candidates produced per generation. Spatial screening with Q-learning-based operator scheduling demonstrated large gains under severely constrained evaluation budgets [1], and sparse adaptive-quota CSO showed that geometric screening of true evaluation slots is the primary performance source with reinforcement learning adding complementary but smaller effects. Surrogate-assisted GWO attacks the same expense asymmetry with models [2]; we deliberately avoid surrogates and keep all

All experiments in this draft come from the frozen batch recorded in the accompanying repository; every number is reproducible from the archived runs.

evaluations true. Surrogate-assisted and asynchronous schemes attack the same asymmetry with models or event-driven evaluation. Our work keeps the sparse-evaluation backbone but asks a different question: given that only K candidates can be evaluated, *how should the pool be composed*? We find that a dual-route pool - refinement plus verified multi-region proposals - matters more than the selection rule, and that geometry-driven selection that ignores the pack's own trajectories can actively hurt.

B. Grey wolf optimizer

Since its introduction [3], GWO has been extended mostly through its search equations or social hierarchy: dimension-wise learning and improved leadership [4], multi-population structures with Laplace mutation [5], strengthened operators with elite re-selection [6], dynamic multi-strategy grouping [7], and reinforcement-learning-based leadership or operator selection [8], [9]. Most relevantly, Q-learning has already been combined with GWO for weighted fitness-distance guidance and for UAV path planning [10], [11]; our study therefore never claims to introduce Q-learning to GWO — it uses a lightweight Q scheduler as one of several pre-registered allocation policies and reports its measured (absence of) benefit. Reviews identify diversity collapse and stagnation as recurring difficulties and note the diminishing returns of mechanism stacking. DRS-GWO keeps the classical update equations and changes only the evaluation interface of the algorithm: which candidates are allowed to exist in the search state, and which of them consume the budget.

C. Lightweight reinforcement-learning schedulers

Small tabular Q-learners have repeatedly been proposed to steer metaheuristic parameters [8], [10], [11]. In our setting the state space is deliberately tiny (three progress levels $\times$ two stagnation levels) and the action set is the fraction of evaluation slots given to the multi-region route, $\{0.25, 0.50, 0.75\}$. The empirical result - the scheduler beats random quota perturbation but not the best fixed quota - is consistent with earlier observations that the gains of such schedulers are secondary to the screening they accompany.

III. Proposed Method

A. Verified-state discipline

The population consists of $N = 50$ verified wolves: each wolf i holds a position x_i and the true fitness value $f(x_i)$ obtained when that exact position was evaluated. Two rules are enforced by construction and by unit tests: (i) a candidate that has not been truly evaluated never enters any fitness ranking, leader update or archive; (ii) accepted positions and fitness values are committed atomically. The leader hierarchy α, β, δ is refreshed only from truly evaluated positions, and the strict historical best is kept for reporting and reward computation.

B. Dual-route candidate pool

Each generation generates $M = 100 = 2N$ cheap candidates, none of which consumes budget:

- **Refinement route ($G_1 + G_2$, 62 candidates).** $N = 50$ standard GWO leader-aggregation moves, one per wolf, using per-dimension random coefficients and the classical linear decay of a over the true budget; plus 12 shell samples $\alpha + \sigma(2\mathbf{u} - 1)$ around the incumbent, where σ is the median length of the 25 most recent *successful* displacements (floored at 1% of the search-space diagonal).
- **Verified multi-region route (E, 38 candidates).** An elite archive (capacity 24) stores evaluated solutions that are strictly better than the archive's worst member and separated from all members by at least 2% of the diagonal. Archive members accumulate *credit* whenever an evaluation in their basin improves a verified record. E candidates are drawn as single-centre shells around credit-weighted archive members, and as uniform points on the segment between two distinct archive centres with a small jitter. The archive is a memory of verified basins; it never contains an unevaluated candidate.

C. Evaluation allocation

Of the $2N$ candidates, exactly $K = 10$ receive the true evaluations of the generation (the frozen value; $K \in \{10, 25, 50\}$ was screened on the development set). The frozen policy selects the K candidates uniformly at random over the pool. Unless selected, a candidate is discarded at the end of the generation: the wolf keeps its verified state, so the pack advances only through evaluations that actually happened. The route mix of the evaluated slots therefore follows the pool proportions in expectation ($\approx 50\%$ refinement moves, 12% incumbent shells, 38% multi-region proposals).

D. Selection and commit

Selected candidates are evaluated in a deterministic order. A selected refinement move commits to its parent wolf; a selected multi-region or shell candidate replaces the worst verified wolf only when strictly better; accepted evaluations update the leader hierarchy immediately and may update the archive. Because every generation consumes exactly K true evaluations, the total budget is exact: N initial evaluations plus B/K generations reaches $B = 100{,}000$ with $50 + 9{,}995 \times 10$ evaluations.

E. Cost

The method is evaluation-neutral: it never spends more true evaluations than GWO (both consume exactly B). The additional arithmetic is the generation of $2N$ cheap candidates and $O(N)$ bookkeeping per generation; on CEC2017 at $D = 50$, 100,000 evaluations, the measured wall-clock is 5.93 seconds per run versus 3.63 for GWO on the same machine and four workers ($K = 10$ performs about ten times more generations, hence the $1.6\times$ factor). Because $K < N$, the pack updates a random subset of wolves per generation and performs 10,000 generations per run instead of 2,000; the finer granularity

Algorithm 1 DRS-GWO (one generation, exactly K true evaluations)

Generate G_1 (one GWO move per wolf) from verified leaders.
Generate G_2 (incumbent shells, success-scale σ).
Generate E (archive-centre shells and dual-centre segments).
Select K candidates uniformly at random from $G_1 \cup G_2 \cup E$.
for each selected candidate c (in index order) **do**
    $v \leftarrow f(c)$ ▷ true evaluation; $+1$ FE
    if c is a refinement move of wolf i **then**
        commit (c, v) to wolf i; update record and σ-ring
    else if $v < \max_j f(x_j)$ **then**
        replace the worst verified wolf by (c, v)
    end if
    refresh α, β, δ; try archive insert (verified only)
end for

is one of the mechanisms producing the early-budget gains reported in Section V.

IV. Experimental Setup

A. Protocol

All experiments use CEC2017 functions F_1 and F_3–F_{30} (F_2 excluded by convention), $D = 50$, and an exact budget of $B = 100{,}000$ true objective evaluations including initialization, with 30 independent runs (seeds 3001–3030). Every objective call of every algorithm passes through a common probe that counts evaluations independently; a run fails if the probe disagrees with the algorithm's counter. The initial population of each (function, seed) pair is drawn once and shared as a prefix by all algorithms, so paired comparisons start from identical points. All runs execute on four local MATLAB workers (Parallel Computing Toolbox, `parfor` over independent algorithm×function×seed jobs); every finished run is written atomically to disk and the pipeline is resumable. No serial fallback and no uncounted evaluations occur anywhere.

B. Development and final isolation

Design decisions (pool composition, K, selection and quota policies, Q scheduler parameters) were made on a development set - eight stratified functions, $D = 30$, 30,000 evaluations, seeds 3001–3010 plus an unseen confirmation set 3011–3020 - under pre-registered gates (accompanying repository, `docs/01–docs/07`). The formal tier at $D = 50$ was executed once, after freezing the method.

C. Baselines and statistics

Strong baselines are reused from audited 30-run tables only after this project reproduced their implementations bit-for-bit under the same protocol (IGWO, HIDMS-PSO, HCLPSO, DMS-PSO, SAQ-CSO, SPSO-2011, HPSO-TVAC, CLPSO, original CSO, and the sparse-evaluation RASE-GWO as an external historical reference). Comparisons use paired Wilcoxon signed-rank tests on $\log_{10}$ errors with Holm correction across the 29 functions, average ranks, and median paired error ratios; behaviour metrics (useful-novelty rate, improving-evaluation rate, stagnation-recovery probability, long- stagnation share) are computed from the same probes for all methods that carry instrumentation.

TABLE I
CEC2017 $D = 50$, 100,000 true evaluations, 29 functions, 30 runs. "W-L vs GWO" counts function-mean wins and losses. Baselines marked "audited" are read-only results whose implementations this project reproduced bit-for-bit before reuse.

Method	Mean rank	W-L vs GWO	Med. ratio vs GWO	Time (s)	Source
HIDMS-PSO	3.72	29-0	0.430	3.32	audited
DMS-PSO	4.48	28-1	0.429	1.89	audited
HCLPSO	4.66	29-0	0.485	2.13	audited
S-star	5.97	28-1	0.525	5.93	this work
S1-pool38	6.52	28-1	0.541	3.30	this work (K=50)
S3p-q25	6.69	27-2	0.548	3.33	this work (quota)
IGWO	7.31	27-2	0.526	2.32	audited
SAQ-CSO	7.69	29-0	0.615	9.08	audited
SPSO-2011	8.24	26-3	0.615	2.56	audited
S5p-q	9.03	27-2	0.624	3.35	this work (Q)
HPSO-TVAC	10.17	17-12	0.931	1.80	audited
B0-ANGWO	11.17	21-8	0.876	3.76	frozen reuse
CLPSO	11.38	21-8	0.791	2.41	audited
RASE-GWO	12.28	25-4	0.913	6.27	audited (historical)
B1-dense-sync	13.14	21-8	0.992	2.88	this work (blank)
GWO	13.59	–	1.000	3.63	audited
Original-CSO	16.97	0-29	3.382	2.24	audited

V. Results

A. Formal comparison

All formal results use $D = 50$, 100,000 true evaluations, 29 functions and 30 runs. Table I reports the full comparison (17 methods). The frozen method (S-star) attains mean rank 5.97, behind the three multi-swarm PSO variants (HIDMS-PSO 3.72, DMS-PSO 4.48, HCLPSO 4.66) but *ahead of the author-faithful I-GWO (7.31) and of sparse adaptive-quota CSO (SAQ-CSO, 7.69)*, and far ahead of standard GWO (13.59), the frozen full-evaluation ANGWO (11.17), the historical sparse-evaluation RASE-GWO (12.28), CLPSO (11.38) and the original CSO (16.97).

Against standard GWO the frozen method wins 28 of 29 functions with a median error ratio of 0.525 and 24 Holm-significant improvements with none significantly worse (pooled $p \approx 7 \times 10^{-103}$). Against the frozen full-evaluation ANGWO it wins 27/29 (ratio 0.723, Holm $+22/-1$). Against I-GWO the comparison is a statistical tie with a positive median: ratio 0.938, 17 wins against 12 losses, pooled $p = 0.55$, and 12 Holm-significant wins against 6 significant losses — large gains on some landscapes (e.g. 7× on F1, 140× on F9, 12× on F13) traded for losses on others (F19 36×, F30 8.6×, F6 and F12/F14 about 2×). Against SAQ-CSO the median ratio is 1.010 (13/16, Holm $+9/-10$), i.e. practical parity with the sparse-evaluation CSO line. Table II aggregates by function family.

B. Early-budget behaviour

At 10% and 20% of the budget the frozen method is already ahead of both GWO (ratios 0.802 and 0.777; 25 and 26 function wins of 29) and the frozen ANGWO (0.732 and 0.768; 27 wins), and the advantage grows to a 0.726 median ratio against

TABLE II
MEDIAN PAIRED ERROR RATIO OF THE FROZEN METHOD OVER EACH BASELINE, AGGREGATED BY FAMILY (MEDIAN OF THE FAMILY'S PER-FUNCTION MEDIANS).

Family	GWO	I-GWO	HIDMS-PSO	HCLPSO
Unimodal (F1,F3)	0.141	0.432	286	230
Simple multimodal (F4–F10)	0.407	0.796	0.998	1.029
Hybrid (F11–F20)	0.407	1.184	5.651	2.699
Composition (F21–F30)	0.733	0.932	1.089	0.946

TABLE III
BEHAVIOUR METRICS ON THE FORMAL TIER (PROBE-INSTRUMENTED METHODS).

Method	Recovery prob.	Long-stag. share	Useful novelty	Nov. rate Q4	Impr. rate Q4
S-star	0.823	0.813	0.00640	0.712	0.00308
S1-pool38	0.816	0.812	0.00590	0.695	0.00266
S3p-q25	0.808	0.821	0.00563	0.745	0.00272
S5p-q	0.780	0.835	0.00629	0.716	0.00264
B0-ANGWO	0.768	0.831	0.00578	0.690	0.00447
GWO	0.757	0.844	0.00347	0.588	0.00481

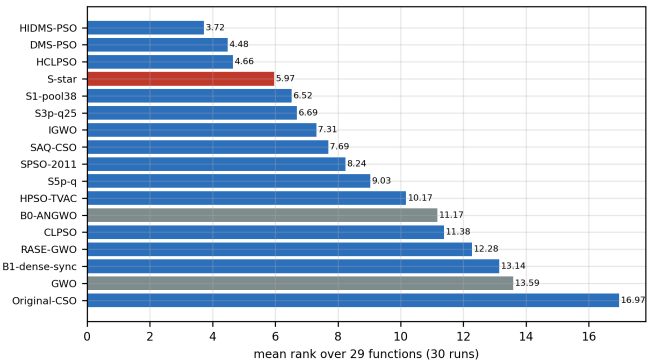


Fig. 1. Mean ranks over the 29 functions (30 runs each), CEC2017 $D = 50$, 100,000 true evaluations. The frozen method is highlighted.

GWO at half budget. The finer evaluation granularity ($K = 10$, hence 10,000 generations) is the mechanism behind the early gains: the pack is updated in ten-evaluation increments instead of fifty-evaluation generations, so the whole budget remains usable for adaptive movement.

C. Behaviour metrics

Table III reports the probe-based behaviour metrics. The frozen method recovers from stagnation episodes more often than both GWO and the frozen ANGWO (0.823 versus 0.757 and 0.768), spends less of the budget in long stagnation (0.813 versus 0.844 and 0.831), and roughly doubles the fraction of its evaluations that convert into improvements after being novel (useful novelty 0.0064 versus 0.0035 for GWO). The improving-evaluation rate itself is *lower* (0.0031 versus 0.0048 per evaluation), which we read as fewer, better-placed discoveries rather than more numerous ones — the earlier-budget and better-recovery numbers show where those discoveries land.

D. Scalability and resource tiers

Two further tiers are reported in the repository. At $D = 100$ (29 functions, 10 seeds, 100,000 evaluations) the frozen method again ranks first among the five methods compared on the overlapping subset (mean rank 1.72 versus S1-K50 2.00, RASE-GWO 2.33, dense-sync 3.52 and I-GWO 3.67), with a 0.703 median ratio and 10 wins against 2 losses (Holm $+5/-1$) against I-GWO on its 12-function subset, at a cost of 11.97 s per run. Under a severely constrained budget (20,000 evaluations, 29 functions, 10 seeds) the method ranks first as well (1.97 versus 2.31 for S1-K50 and 3.83 for GWO), winning 24 of 29 functions against GWO (ratio 0.721) and 24/29 against the frozen ANGWO (0.807). The sparse framework therefore scales with dimension and with smaller budgets rather than relying on the 100,000-evaluation regime.

VI. Ablation and Mechanism Study

The design space was explored on an eight-function development set ($D = 30$, 30,000 evaluations, seeds 3001–3010 plus an unseen confirmation set 3011–3020) under pre-registered gates; every rejected alternative is retained. Unless stated, ratios compare paired runs on identical (function, seed) pairs and a ratio below one favours the variant.

A. Pool composition and evaluation density

Enlarging the multi-region share beyond the pool proportion hurts ($q_E = 0.50$: 1.06; $q_E = 0.75$: 1.67 versus uniform selection); shrinking it does not help (pinned 25% quota: 1.02–1.09 across two seed sets). The uniform policy over the pool is therefore not a compromise but the optimum of the tested family. For the evaluation density, $K = 10$ beat $K = 50$ consistently on both development seed sets (0.970 with 6/8 wins, and 0.955 with 6/8 wins, $p = 0.053$), which motivated the frozen $K = 10$; the formal tier confirms the choice (S-star 0.525 versus S1-pool38 0.541 against GWO, with the earlier-budget advantage).

B. Screening rules are falsified

Directed screening (nearest-incumbent refinement slots plus credit-weighted multi-region centres) is worse than uniform random selection of the same pool (median ratio 1.39); history-informed assignment inside stratified slots is worse still (1.53). The mechanism is visible in the first-generation composition audit: uniform selection evaluates $[G_1, G_2, E] = [24, 10, 16]$ candidates on average, identical in expectation to the stratified allocation, so the loss cannot come from the quota — it comes from starving the pack's own leader-aggregation moves of evaluations, after which the verified population stops advancing.

C. Quota policies and Q-learning

Fixed quotas away from the pool proportion and randomly perturbed quotas both degrade performance (random quota: 1/7 function wins, $p < 10^{-3}$). Three Q-learning schedulers were tested with the pre-registered 6-state (progress $\times$ stagnation) by 3-action (exploration share $\{0.25, 0.50, 0.75\}$) design and

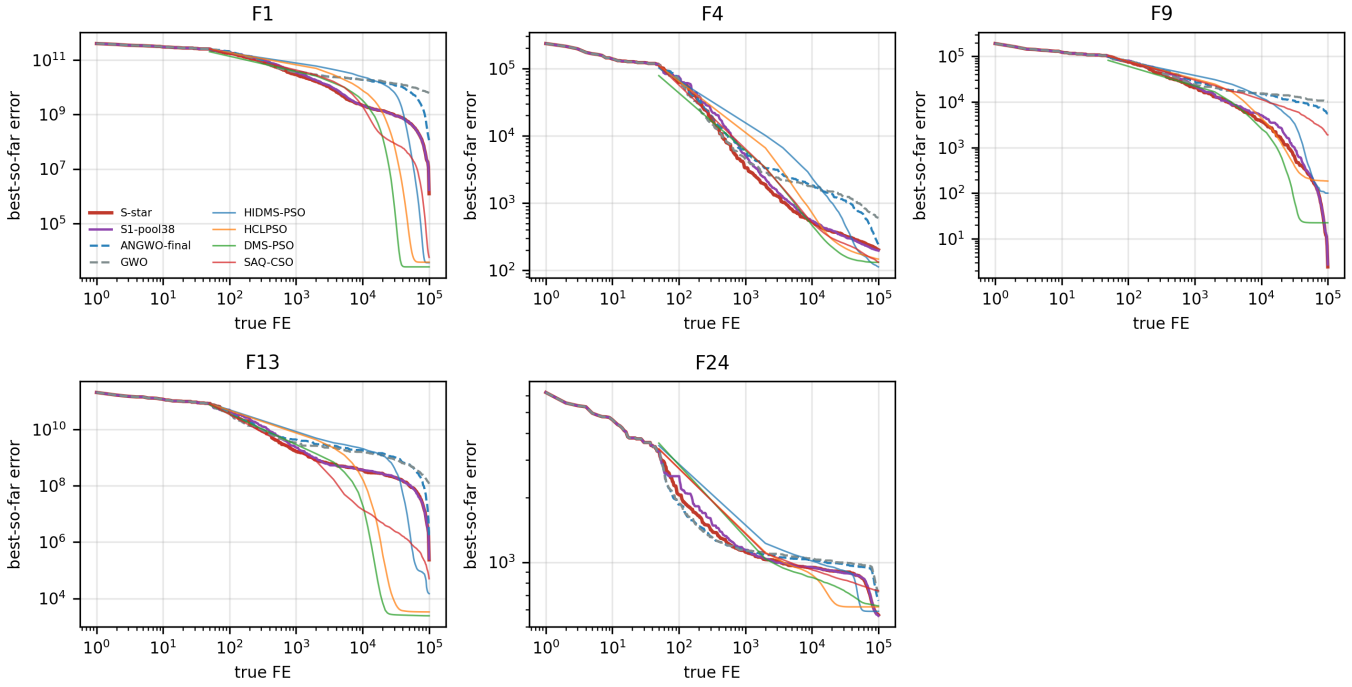


Fig. 2. Median best-so-far error versus true evaluations on five representative functions (F1, F4, F9, F13, F24). Thick red: frozen method; purple: its K=50 variant; dashed grey: standard GWO; dashed blue: frozen full-evaluation ANGWO; thin lines: audited strong baselines. All curves use the same true-FE budget.

per-500-FE reward windows combining the normalized global-best improvement with a small verified-record term: (i) zero-initialized table, (ii) tuned $\alpha = 0.3$, $\gamma = 0.95$, fast ε-decay, and (iii) prior-initialized table. Their ratios against the best fixed quota at 10 development seeds were 0.988, 1.024 and 1.130 (win rates 3/8, 2/8, 4/8); in the formal tier the Q variant (S5p-q) ranks 9.03 against the fixed-quota variant's 6.69. The schedulers do beat *random* quota perturbation (0.75–0.87), but never the best constant; the prior-initialized variant degenerates to 93% single-action play. We therefore conclude that the lightweight Q scheduler has no independent benefit in this setting, and the frozen method contains no RL component — an outcome consistent with earlier findings that the screening term, not the scheduler, carries such frameworks.

D. Post-freeze enhancement probes

Three additional single-change enhancements were screened after the frozen batch: (i) stagnation-triggered radius inflation of the shell candidates ($\times 3$): 1.20, rejected; (ii) reuse of *verified success directions* — 25% of the refinement moves are extended by half a median successful step along a recorded improving displacement: 0.934 (5/8), a passing but small effect; (iii) coordinate-subset perturbation of the incumbent shells (20% of dimensions): error-neutral (0.991) but with a large stagnation-axis gain (recovery 0.887 versus 0.815, long-stagnation share 0.785 versus 0.852, useful novelty $+28\%$). The combination and a fresh-seed confirmation are reported in the accompanying repository; they are deliberately kept outside the frozen claim.

VII. Discussion

A. Dual-axis verdict

The pre-registered gates separate two claims. *Exploration efficiency (A): supported.* At 20% of the budget the frozen method is 23% better than the frozen ANGWO on median (0.768, 26/29 wins), 29% better on the simple-multimodal family (0.714), and its useful-novelty rate is 11% higher (0.00640 versus 0.00578). *Stagnation recovery (B): partially supported.* The recovery probability rises by 5.6 percentage points (0.823 versus 0.768; gate $+0.05$ passed) and the long-stagnation share falls by 1.8 points (0.813 versus 0.831; the pre-registered 0.03 gate is *not* passed). We therefore claim axis A and report axis B as partially supported, with the failed half stated explicitly.

B. Why the simple policy wins

Three independent comparisons point the same way: uniform random selection over a dual-route pool beats geometry-directed screening (1.39), pinned stratified allocation (1.02–1.09), random quota perturbation (1/7 wins) and learned quota scheduling (0.99–1.13 against the best constant). The interpretation is structural, not statistical: in a leader-driven method, the dominant information channel is that each wolf's own move receives real evaluations; screening or scheduling that concentrates evaluation slots away from this channel starves the population, while the dual-route pool itself continuously injects verified multi-region candidates. This is consistent with the earlier observation that evaluations near the incumbent convert an order of magnitude more often than far-away ones.

C. Limitations

(i) The method remains behind the three strongest multi-swarm PSO variants (median ratios 1.04–1.23; losses on F1 by three orders of magnitude on the ill-conditioned unimodal function, and on F6/F19/F30); it ties rather than beats I-GWO and SAQ-CSO. (ii) The frozen $K = 10$ costs about $1.6\times$ GWO's wall-clock (5.93 s versus 3.63 s on four workers) and $2.6\times$ I-GWO's. (iii) The Q-learning component — a headline of the investigated direction — is reported as *not beneficial*; we keep it as a fully documented negative result rather than claiming RL. (iv) The study is CEC2017-only; the $D = 100$ tier and a 20,000-evaluation resource tier are reported separately in the repository, and no engineering application is included. (v) The post-freeze enhancements (success-direction reuse, coordinate-subset shells) show promise on single axes but are not part of the frozen claim.

VIII. Conclusion

This paper studied what a grey wolf pack should spend its scarce true evaluations on. A minimal dual-route sparse framework — refinement candidates near the incumbent plus verified multi-region candidates from an elite archive, with $K = 10$ uniformly sampled evaluations per generation and a strict verified-state discipline — improves standard GWO by a median factor of 0.525 with 24 Holm-significant wins and no significant losses, overtakes the frozen full-evaluation ANGWO and the historical sparse RASE-GWO, and reaches statistical parity with author-faithful I-GWO and with sparse adaptive-quota CSO while remaining behind the strongest multi-swarm PSO variants. A complete pre-registered ablation shows that directed screening, stratified allocation, fixed quota deviation, random quota perturbation and three Q-learning schedulers all fail to beat the simple policy; the exploration-efficiency axis is supported, the stagnation recovery axis is partially supported, and the RL component is reported as an honest negative result.

References

[1] A. manuscript (user provided), "Spatial screening and reinforcement learning-based particle swarm optimization for real-time or resource-constrained engineering optimization," *Under review*, 2026.

[2] H. Dong and Z. Dong, "Surrogate-assisted grey wolf optimization for high-dimensional, computationally expensive black-box problems," *Swarm and Evolutionary Computation*, 2020.

[3] S. Mirjalili, S. M. Mirjalili, and A. Lewis, "Grey wolf optimizer," *Advances in Engineering Software*, vol. 69, pp. 46–61, 2014.

[4] M. H. Nadimi-Shahraki, S. Taghian, and S. Mirjalili, "An improved grey wolf optimizer for solving engineering problems," *Expert Systems with Applications*, vol. 166, p. 113917, 2021.

[5] Z. Wang, L. Shu, and S. Yang, "Multi-population dynamic grey wolf optimizer based on dimension learning and laplace mutation for global optimization," *Expert Systems with Applications*, 2025.

[6] X. Chen, C. Ye, and Y. Zhang, "Strengthened grey wolf optimization algorithms for numerical optimization tasks and automl," *Swarm and Evolutionary Computation*, 2025.

[7] Y. Wang, J. Song, and H. Zhao, "Dynamic multi-strategy grey wolf optimizer and its applications," *Scientific Reports*, 2026.

[8] A. Maleki, M. Roayaei, and S. Mirjalili, "Enhancing leadership-based metaheuristics using reinforcement learning: a case study in grey wolf optimizer," *Knowledge-Based Systems*, 2025.

[9] G. Yin, W. Li, and Q. Xiong, "Adaptive multi-strategy grey wolf optimizer with reinforcement learning for multi-objective precision irrigation optimization," *Algorithms*, 2026.

[10] C. Huang, B. Yang, and Y. Huo, "QLFDGWO: Q-learning-guided weighted fitness-distance grey wolf optimizer for UAV path planning," *Biomimetics*, 2026.

[11] G. Nayeem, M. Fan, and G. Daiyan, "Adaptive q-learning grey wolf optimizer for UAV path planning," *Drones*, 2025.